MAFFT command-line options

All options taken by the `mafft` command. The list comes from the option parser in the `mafft` script (`core/mafft.tmpl`) of MAFFT v7.471 (July 2020). The four options added later, up to v7.511, come from the changelog. The current source (v7.526) wasn't available to check; see the notes at the end.

The tag after each option shows where it is documented, using the keys in the URL list at the bottom. [src] means the option appears only in the source code. Its description is inferred from what the code does, so treat it as approximate.

Help and version

- `--auto`: choose L-INS-i, FFT-NS-i or FFT-NS-2 automatically from data size. [MAN, ADD, TREEIN]
- `--help`, `--info`: print a short usage summary. [src]
- `--version`: print the version and exit. [src]
- `--man`: show the manual page. [src]

Pairwise / distance stage (strategy)

- `--6merpair`: distances from shared 6-mers; the default for FFT-NS methods. [MAN, ADD, VIRAL, REGION, SRV-FRAG]
- `--10merpair`: the same, using 10-mers. [src]
- `--6mermultipair`, `--10mermultipair`: k-mer distances, "multi" variant. [src]
- `--multipair`: accurate (costly) pairwise stage for `--addfragments`. [ADD, SRV-FRAG]
- `--globalpair`: all-to-all Needleman–Wunsch; with `--maxiterate 1000` this is G-INS-i. [MAN, ALG, TREEOUT, MT]
- `--localpair`: all-to-all Smith–Waterman; with `--maxiterate 1000` this is L-INS-i. [MAN, ALG, TREEOUT, DASH]
- `--genafpair`: local alignment with generalized affine gaps; with `--maxiterate 1000` this is E-INS-i. [MAN, ALG]
- `--localgenafpair`: alias for `--genafpair`. [src]
- `--oldgenafpair`: E-INS-i with the parameters used before v7.243. [ALG, MPI]
- `--globalgenafpair`: removed; exits with "no longer supported". [src]
- `--fastapair`: pairwise alignments by FASTA (FASTA must be installed). [MAN]
- `--fastswpair`: FASTA in Smith–Waterman mode. [MAN]
- `--blastpair`: pairwise alignments by BLAST. [MAN (named under `--weight1` only)]
- `--lastpair`, `--lastmultipair`: pairwise alignments by LAST. [src; see Katoh & Frith 2012]
- `--last_m n`, `--last_e n`: LAST tuning parameters passed to the pairwise step. [src]
- `--last_subopt`, `--last_once`: further LAST-mode switches. [src]
- `--hybridpair`: hybrid distance mode. [src]
- `--shortlongpair`, `--longshortpair`: local pairwise mode with zeroed offsets; a source comment says it is used with `--addfull`. [src]

RNA structural alignment (Q-INS-i / X-INS-i)

- `--scarnapair`: MXSCARNA pairwise structural alignment (default for X-INS-i). [NCRNA]
- `--larapair`: LaRA pairwise structural alignment. [NCRNA]